

Chapter 10

Water Resources

Class 10 - Total Geography Morning Star

Choose the correct option

Question 1

What is an aquifer?

1. An underground well that readily gives out water on digging the land.
2. An underground pond that was once a source of water and is now buried under rocks.
3. An underground stream saturated with water that is transmitted readily on to the surface.
4. None of the above.

Answer

An underground stream saturated with water that is transmitted readily on to the surface.

Question 2

Which of the following measures are not used to conserve water?

1. Rainwater harvesting
2. Using bottled drinking water only
3. Water recycling
4. Preventing water pollution

Answer

Using bottled drinking water only

Question 3

Which of the following is not a secondary source of water?

1. Rivers
2. Groundwater

3. Lakes and ponds
4. Rain

Answer

Rain

Question 4

Sprinkler irrigation is also known as

1. Seepage irrigation
2. Overhead irrigation
3. Protected irrigation
4. Nozzle irrigation

Answer

Overhead irrigation

Question 5

Tanks form the main source of irrigation in

1. Punjab and Haryana
2. the Deccan Plateau
3. Madhya Pradesh and Gujarat
4. All of the above.

Answer

the Deccan Plateau

Question 6

..... is not a conventional means of irrigation.

1. Furrow irrigation
2. Wells
3. Tanks
4. Inundation canals

Answer

Furrow irrigation

Question 7

The most advanced and efficient method of irrigation is

1. Tank irrigation.
2. Drip irrigation.
3. Canal irrigation.
4. Well irrigation.

Answer

Drip irrigation

Question 8

Most of the canals in India belong to which category?

1. Perennial canals
2. Inundation canals
3. Navigation canals
4. None of the above.

Answer

Perennial canals

Question 9

Which of the following least irrigated state of India is solely dependent on canals?

1. Assam
2. Mizoram
3. Tripura
4. Arunachal Pradesh

Answer

Mizoram

Question 10

Which of the following is the most advanced and efficient method of irrigation?

1. Furrow irrigation
2. Spray irrigation
3. Drip irrigation
4. Sprinkler irrigation

Answer

Drip irrigation

Answer the following questions

Question 1(a)

What do you understand by the term 'water resource'?

Answer

The term 'water resource' refers to any of the entire range of natural waters that occur on the earth and are of potential use to living beings.

Question 1(b)

Give two points of difference between 'surface water' and 'ground water'.

Answer

Surface water	Ground water
Surface water forms streams, lakes, rivers and ponds flowing on land.	The water which seeps through the cracks and crevices in the land is known as ground water.
It can be used directly without pumping.	It needs to be pumped out of the ground before it can be used.

Question 1(c)

Give a geographical reason for each of the following:

- (i) Need to adopt different means of irrigation.
- (ii) Need for conserving water.
- (iii) Man is responsible for water crisis in India.

Answer

(i) There is a need to adopt different means of irrigation because of uncertainty of rainfall, uneven distribution of rainfall, crop requirements, nature of the soil, effective utilisation of river water and to maximise production.

(ii) There is a need for conserving water for the following reasons-

1. The over exploitation of ground water often results in the lowering of water table.
2. The loss of vegetation causes drought and reduction of rainfall and lowering of the water table.
3. Irrigation utilises more than 90% of the total freshwater.
4. The increase in population results in water scarcity.
5. Our water resources are polluted and their water can hardly be used without adequate treatment.

(iii) Man is responsible for water crisis in India because due to increase in population, irrigation and industrialisation, the demand for water has risen. This has led to a decline in groundwater levels in various parts of the country.

Question 1(d)

(i) What is meant by rainwater harvesting?

(ii) Mention any two rainwater harvesting systems practised in India.

Answer

(i) Rainwater harvesting is the process of increasing the recharge of groundwater by capturing and storing rainwater locally in sub-surface water reservoirs.

(ii) Two rainwater harvesting systems practised in India are-

1. Khatri in western Himalayas
2. Johads in central India

Question 2(a)

What is meant by the term irrigation?

Answer

Irrigation refers to the process of watering of agricultural plants through artificial means from wells, tanks, tube wells, canals, etc.

Question 2(b)

What is meant by the term 'water scarcity'? What has caused this scarcity in India?

Answer

Water scarcity is the lack of sufficient available water resources to meet the demands of water usage within a region.

Growing population, over-exploitation and unequal distribution of water among social groups are the main causes of water scarcity in India.

Question 2(c)

Give a geographical reason for each of the following:

- (i) Inundation canals are being converted to perennial canals.
- (ii) Tank irrigation is preferred over other means of irrigation in Peninsular India.
- (iii) Groundwater reserves are depleting at a fast rate.

Answer

(i) Inundation canals are being converted to perennial canals because inundation canals get supply of water only when the rivers are in flood while perennial canals can draw water throughout the year and irrigate large areas.

(ii) Tank irrigation is preferred over other means of irrigation in Peninsular India because-

1. The rivers of Deccan are mainly dependent on the rainwater.
2. Many streams become torrential during the rainy season but dry up when the rain ceases.
3. The hard rocks in the area do not absorb waters, wells cannot be made there.
4. The terrain is uneven with many natural depressions where tanks can be easily built.

(iii) Groundwater reserves are depleting at a fast rate due to increase in population, irrigation and industrialisation, the demand for water has risen. This has led to a decline in groundwater level in various parts of the country.

Question 2(d)

- (i) What is meant by traditional or conventional methods of irrigation?
- (ii) Name any two conventional methods of irrigation.

Answer

(i) Traditional methods of irrigation means to supply water to the plant zone by pumped water from surface or subsurface sources like ponds, rivers, channels or groundwater through earthen channels or pipes with gravitational force.

(ii) Two conventional methods of irrigation are wells and tanks.

Question 3(a)

State any two drawbacks of conventional methods of irrigation.

Answer

Two drawbacks of conventional methods of irrigation are-

1. A large quantity of water is not properly utilised.
2. The fields situated in low areas always get excess water causing prolonged water logging.

Question 3(b)

Give two advantages and two disadvantages of well irrigation.

Answer

Two advantages of well irrigation are-

1. Wells can be dug at a very low cost which is well within the means of poor farmers.
2. By the use of pumps and tubewells, water can be lifted even from great depths.

Two disadvantages of well irrigation are-

1. Wells depend on underground water resources whose distribution varies from region to region.
2. The traditional wells dry up due to over withdrawal of the ground water and lowering of the water table.

Question 3(c)

Give a geographical reason for each of the following:

- (i) Well irrigation is confined mainly to the alluvial plains.
- (ii) In Tamil Nadu, nearly one-third of the net irrigated area is under canal irrigation.
- (iii) Drip irrigation reduces loss of water through evaporation.

Answer

(i) Well irrigation is confined mainly to the alluvial plains as owing to the soft nature of the soil, wells can be easily dug and the yield of crops from the land after irrigation is considerably high.

(ii) In Tamil Nadu, nearly one-third of the net irrigated area is under canal irrigation because Tamil Nadu receives rainfall during the winter season while summer season remains dry. Thus, irrigation is needed in summer to make up the lack of rainfall.

(iii) Drip irrigation reduces loss of water through evaporation as this system consists of perforated pipes that are placed between rows of crops or buried along their root lines and give water directly on to the crops.

Question 3(d)

(i) Name any two states where well irrigation is practised.

(ii) Give one advantage and one disadvantage of tubewell irrigation.

Answer

(i) Two states where well irrigation is practised are Punjab and Uttar Pradesh.

(ii) One advantage of tubewell irrigation is that it brings up clean water.
One disadvantage of tubewell irrigation is that it is useless if the water is brackish.

Question 4(a)

Name the two types of canals. Name two states where perennial canals are widely used.

Answer

Two types of canals are-

1. Inundation canals
2. Perennial canals

Two states where perennial canals are widely used are Uttar Pradesh and Madhya Pradesh.

Question 4(b)

How are the fields irrigated using the Persian wheel method?

Answer

Persian wheel method is a partly submerged vertical wheel with buckets attached to the rim. As the wheel is turned by draught animals rotating a geared horizontal wheel, the buckets are filled and emptied into a trough above, which carries the water to crop-sown fields.

Question 4(c)

Give a geographical reason for each of the following:

- (i) Canals make the soil infertile.
- (ii) Tubewell irrigation is quite expensive.
- (iii) Excessive accumulation of salts make the soils unsuitable for cultivation.

Answer

- (i) Canals make the soil infertile because in canal irrigation, where water table is a few feet below the ground, the alkaline salts come to the surface, mix with the soil and make it unproductive.
- (ii) Tubewell irrigation is quite expensive as it requires continuous supply of electricity.
- (iii) Excessive accumulation of salts make the soils unsuitable for cultivation because when salt concentrations in the soil are high, the movement of water from the soil to the root is slowed down. When the salt concentrations in the soil are higher than inside the root cells, the soil will draw water from the root, and the plant will wilt and die. Hence, the soil becomes unsuitable for cultivation.

Question 4(d)

- (i) Name two states in which tubewells are extensively used.
- (ii) State why tubewell irrigation is important in Punjab.

Answer

- (i) Two states in which tubewells are extensively used are Punjab and Haryana.
- (ii) Tubewell irrigation is important in Punjab because it irrigates large areas of land in comparatively less time and it can irrigate the fields throughout the year.

Question 5(a)

What is meant by rainwater harvesting?

Answer

Rainwater harvesting is the process of increasing the recharge of groundwater by capturing and storing rainwater locally in sub-surface water reservoirs.

Question 5(b)

State any two methods of rainwater harvesting.

Answer

Two methods of rainwater harvesting are-

1. Khatri in western Himalayas
2. Johads in central India

Question 5(c)

Give a geographical reason for each of the following:

- (i) Many farmers in India still use wells.
- (ii) There is very little recharge of groundwater.
- (iii) The traditional wells dry up.

Answer

- (i) Many farmers in India still use wells because wells can be dug at a very low cost and the oxen which are kept for ploughing the land can be utilised for drawing water from the well.
- (ii) There is very little recharge of groundwater as only a small amount of rain water runs down the earth surface and gets stored in the form of groundwater. So, only a small percentage of water gets down to the water table and forms ground water.
- (iii) The traditional wells dry up due to the over-withdrawal of the groundwater and lowering of the water table.

Question 5(d)

- (i) What is watershed management?
- (ii) How is it beneficial for farmers in the long run?

Answer

- (i) Watershed management refers to the efficient management and conservation of both the surface and groundwater resources. It includes the prevention of run-off as well as storage and recharge of groundwater by various methods like percolation pits, recharge wells, borewells, dugwells etc.
- (ii) Watershed management is beneficial for farmers as it is aimed at conserving both soil and water. Since both soil and water are indispensable for the cultivation of crops, watershed management helps farmers to maximize their crop production and income.

Question 6(a)

What is 'drip irrigation'? How is it useful?

Answer

Drip irrigation is the most advanced and efficient method of irrigation. This system consists of perforated pipes that are placed between rows of crops or buried along their root lines and give water directly on to the crops.

It is useful as it reduces evaporation drastically and irrigation water is conserved. It also allows the grower to customise an irrigation programme most beneficial to each crop.

Question 6(b)

Explain briefly the need to conserve water.

Answer

We need to conserve water for the following reasons-

1. The over exploitation of ground water often results in the lowering of water table.
2. The loss of vegetation causes drought and reduction of rainfall and lowering of the water table.
3. Irrigation utilises more than 90% of the total freshwater.
4. The increase in population results in water scarcity.
5. Our water resources are polluted and their water can hardly be used without adequate treatment.

Question 6(c)

Give a geographical reason for each of the following:

- (i) Modern means of irrigation are gaining popularity.
- (ii) Sprinkler irrigation helps in conserving water.
- (iii) Spray irrigation is quite expensive.

Answer

(i) Modern means of irrigation are gaining popularity because of the following reasons-

1. No loss of water due to seepage or evaporation
2. Conserve water
3. Prevent soil erosion

4. Suitable for areas where the rainfall is low
5. High efficiency

(ii) Sprinkler irrigation helps in conserving water as water is sprayed directly to the required plants so there is no loss of water through seepage or evaporation.

(iii) Spray irrigation is quite expensive because it requires complex machinery.

Question 6(d)

What is meant by furrow irrigation? What is its advantage?

Answer

Furrow irrigation is a type of flood irrigation in which the water poured on the field is directed to flow through narrow channels dug between the rows of crops, instead of evenly distributing the water throughout the whole field. The furrows must have equal dimensions, in order to guarantee that the water is distributed evenly.

The advantage of furrow irrigation is lower initial investment of equipment and lower pumping costs per acre-inch of water pumped.

Thinking Skills

Question 1

Nature has endowed India with plentiful water resources. Despite this, acute shortage of water is noticed in some states. What could be the reasons for this shortage?

Answer

India is known to be a land of rivers with 12 major rivers and 46 medium rivers. Many of these rivers are perennial and some of them are seasonal. The annual rainfall in India is 1,170 mm which is higher as compared to the world average of 800 mm. Despite this, acute shortage of water in some states is due to the following reasons:

1. **Uneven distribution** — Water resources are not distributed evenly across the country. The monsoons rains are also spread unevenly. The result is that some states are devastated with floods while others face severe drought at the same time.
2. **Population growth** — India has one of the world's largest populations, and the demand for water is increasing due to rapid urbanization, industrialization, and agricultural growth. India constitutes around 17% of the world's population but has only 4% of the world's freshwater resources.

3. **Poor water management** — Inefficient management of water resources, lack of proper infrastructure, and poor governance are also contributing to the water shortage in some states.
4. **Pollution** — Pollution of water bodies, such as rivers, lakes, and groundwater, is reducing the availability of clean water for domestic, agricultural, and industrial use.

Question 2

In your city, critical shortage of water has been reported due to the sharp fall in the groundwater level. What steps you will take to check this scarcity of water?

Answer

To check this scarcity of water, we need to take quick steps to make proper policies and laws and adopt effective measures for water conservation. This can be done by using rainwater harvesting. It is the process of increasing the recharge of groundwater by capturing and storing rainwater locally in sub-surface water reservoirs.

Various kinds of recharge structures are possible which can ensure that rainwater percolates in the ground instead of draining away from the surface. Some of these are:

1. Borewells and Dugwells — They are used to raise the underground water table.
2. Recharge Pit — They are excavated into the ground lined with a brick or stone wall with openings at regular intervals.
3. Percolation Pits — It is a bored hole of up to 30 cm diameter drilled in the ground to a depth of 3 to 10 m.
4. Recharge Trenches — They are excavated on the ground and refilled with porous media like pebbles, boulder or brick bats to harvest the surface runoff.
5. Permeable Surfaces — For example, a patch of grass is used to retain a large proportion of the rainwater falling on it, yielding only 10-15 per cent as runoff.
6. Porous tiles — They can be used on pavements and footpaths.

Question 3

A number of water saving technologies have been developed in recent decades. Which one would you prefer to use in your area and why?

Answer

I would like to use rainwater harvesting technique to conserve water because the area where I reside receives adequate rainfall during the monsoon season. My locality is developed in such a way that every block has a big park with patches of grass and trees. The trees and grass minimize runoff.

The people of my area can collectively harvest rainwater on their rooftops and even use the parks for building various recharge structures such as borewells, recharge pits, percolation pits, recharge trenches, porous tiles on the pavement etc.

Question 4

If you are given a choice between using traditional or modern methods of irrigation, which method you would prefer to use in your area and why?

Answer

I would prefer to use drip irrigation which is a modern method of irrigation because of the following reasons:

1. It is the most advanced and efficient method of irrigation.
2. This system consists of perforated pipes that are placed between rows of crops or buried along their root lines and give water directly on to the crops. As a result, evaporation is drastically reduced.
3. It helps to conserve irrigation water.
4. It allows the grower to customise an irrigation programme most beneficial to each crop.